

i. Why do computers use Cache Memory?

- Computers use cache memory to store frequently accessed data and instructions, providing faster access times compared to main memory (RAM). This helps improve overall system performance by reducing the time the CPU spends waiting for data.

ii. How many bits make a kilobyte?

- A kilobyte consists of 8,192 bits. (Note: 1 kilobyte = 1024 bytes, and each byte is 8 bits.)

iii. Why don't we use Flash memory as RAM?

- Flash memory has limited write cycles and slower write speeds compared to RAM. Using Flash memory as RAM would result in faster wear and tear, reducing its lifespan. Additionally, RAM is designed for frequent read/write operations, which Flash memory is not optimized for.

iv. Why is it hard to make more than 1GB processor in a single chip?

- Increasing the number of transistors and components in a single chip beyond a certain point faces challenges related to heat dissipation, power consumption, and manufacturing complexities. These factors make it technically challenging and economically impractical to create processors with extremely high capacities on a single chip.

v. What are the components that make a computer system?

- The key components of a computer system include:
 - Central Processing Unit (CPU)
 - Memory (RAM and storage)
 - Motherboard
 - Input devices (keyboard, mouse, etc.)
 - Output devices (monitor, printer, etc.)
 - Storage devices (hard drive, SSD, etc.)
 - Power supply
 - Graphics Processing Unit (GPU)
 - Network interface (for connectivity)
 - Operating System (OS)